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Corporate ESG Profiles and Investor Horizons

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1 Big picture

- **Question.** Do longer-horizon institutional investors tilt their portfolios toward firms with better Environmental, Social, and Governance (ESG) profiles? If yes, what mechanism explains the relation?
- **Core idea.** ESG projects, risks, and controversies often matter over long horizons. Investors with longer horizons may therefore have stronger incentives and better ability to evaluate ESG information than short-horizon investors.
- **Empirical setting.** The paper studies U.S. actively managed domestic equity mutual funds and 13f institutional investors from 2000 to 2018, combining portfolio holdings with firm-level ESG ratings.
- **Baseline result.** Long-horizon investors hold portfolios with higher average ESG scores. The pattern appears both at the *investor level* and at the *firm level*: high-ESG firms have shareholder bases with longer average horizons.
- **Text evidence.** Even among funds without an explicit ESG or SRI mandate, longer-horizon mutual funds are more likely to discuss ESG considerations in shareholder reports.
- **Rating robustness.** The result does not rely only on MSCI. It also appears with Refinitiv and Sustainalytics ESG ratings, and remains under an errors-in-variables IV approach that instruments MSCI ESG with the other ratings.
- **Information-channel result.** Long-term funds are more patient after poor earnings news but sell more aggressively after negative environmental or social incidents. This contrast suggests that long- and short-term investors process different types of information differently.
- **Limits-to-arbitrage result.** Long-term funds have lower flow-performance sensitivity, and funds with lower flow-performance sensitivity hold higher-ESG portfolios. A regulatory shock that shortened fund horizons also reduced ESG tilts.
- **Catering result.** The evidence does not support a simple clientele-catering explanation. Long-term-fund investors do not respond more strongly to Morningstar sustainability ratings, and long-term funds do not increase ESG tilts more in high climate-news periods.
- **Broader takeaway.** The paper reframes ESG investing as a horizon-dependent investment behavior. ESG preferences are not just about whether an institution is "pro-ESG." They depend on how long the institution can wait, what information it specializes in, and how exposed it is to short-term flow pressure.

2 Data and measurement (key objects)

2.1 Institutional holdings and sample construction

The paper uses two main holdings databases:

- **Thomson Reuters s12** for mutual fund equity holdings;
- **Thomson Reuters s34** for 13f institutional equity holdings.

Mutual fund characteristics come from the CRSP Mutual Fund database. The paper removes ETFs, index funds, and funds with very broad index-like portfolios, so the baseline fund sample focuses on active U.S. domestic equity mutual funds. It also removes mutual funds with explicit ESG or SRI mandates.

Interpretation. The exclusion of explicit ESG/SRI funds is important. The paper is not merely showing that ESG funds hold ESG stocks. It asks whether ordinary active funds with longer horizons reveal more ESG preference.

2.2 ESG ratings

The main ESG measure comes from MSCI ESG STATS. The measure aggregates positive indicators, or strengths, and negative indicators, or concerns, across environmental, social, and governance categories.

The paper also uses two alternative ESG-rating providers:

- **Sustainalytics**;
- **Refinitiv** (formerly ASSET4).

Interpretation. ESG ratings are noisy and rating agencies disagree. The paper treats this not just as a robustness issue, but also as a measurement-error problem. Consistent results across several rating systems make it less likely that the finding is an artifact of one provider's scoring method.

2.3 Investor horizon: mutual funds

The first mutual fund horizon measure is the fund's reported portfolio turnover ratio. Higher turnover means a shorter implied horizon.

The second measure is the holdings-based churn ratio. For fund j in quarter t,

$$CR_{j,t} = \frac{\sum_{i \in I} \left| Shares_{i,j,t} P_{i,t} - Shares_{i,j,t-1} P_{i,t-1} - Shares_{i,j,t-1} \Delta P_{i,t} \right|}{\sum_{i \in I} \left(Shares_{i,j,t} P_{i,t} + Shares_{i,j,t-1} P_{i,t-1} \right) / 2}. \tag{1}$$

The paper uses a four-quarter moving average of the churn ratio to reduce measurement error.

Interpretation. The numerator removes the mechanical effect of price changes and tries to isolate active trading. A high value means the fund frequently changes its holdings, so it is treated as a shorter-horizon investor.

2.4 Adjusted churn ratio

Fund flows can mechanically raise measured churn because inflows require purchases and outflows require sales. To reduce this problem, the paper calculates separate buy-side and sell-side churn measures:

$$CR^{buy}_{j,t} = \frac{\sum_{i: Shares_{i,j,t} > Shares_{i,j,t-1}} \left| Shares_{i,j,t} P_{i,t} - Shares_{i,j,t-1} P_{i,t-1} - Shares_{i,j,t-1} \Delta P_{i,t} \right|}{\sum_{i \in I} \left(Shares_{i,j,t} P_{i,t} + Shares_{i,j,t-1} P_{i,t-1} \right) / 2}, \tag{2}$$

$$CR_{j,t}^{sell} = \frac{\sum_{i:Shares_{i,j,t}<Shares_{i,j,t-1}} |Shares_{i,j,t}P_{i,t} - Shares_{i,j,t-1}P_{i,t-1} - Shares_{i,j,t-1}\Delta P_{i,t}|}{\sum_{i\in I} (Shares_{i,j,t}P_{i,t} + Shares_{i,j,t-1}P_{i,t-1})/2}. \quad (3)$$

The adjusted churn ratio is

$$CR_{j,t}^{Adj} = \min\{CR_{j,t}^{buy}, CR_{j,t}^{sell}\}. \quad (4)$$

Interpretation. If a fund receives large inflows, purchases may be high even though the manager is not changing the portfolio aggressively. Taking the smaller of buy and sell churn reduces the influence of one-sided flow-driven trading.

2.5 Investor horizon: 13f institutions

13f institutions do not report a CRSP-style portfolio turnover ratio. The paper therefore uses holdings-based churn and adjusted churn as the main horizon proxies for 13f investors.

As an additional measure, the paper uses the Bushee classification:

- **transient investors**: high turnover and short horizon;
- **dedicated investors**: concentrated, long-term investors;
- **quasi-indexers**: diversified, low-turnover investors.

Interpretation. The Bushee measure is useful because it classifies institutions using both turnover and portfolio diversification. If high-ESG firms have fewer transient owners, the result is not tied to one narrow churn formula.

2.6 Firm-level shareholder horizon

For each firm, the paper constructs the average horizon of its institutional shareholder base. For example, the firm-level mutual fund turnover measure is

$$FirmLevelTurnover_{i,t} = \frac{\sum_{j\in J} Turnover_{j,t} \times Shares_{i,j,t}}{\sum_{j\in J} Shares_{i,j,t}}. \quad (5)$$

Similarly, the firm-level churn ratio is

$$FirmLevelChurn_{i,t} = \frac{\sum_{j\in J} CR_{j,t} \times Shares_{i,j,t}}{\sum_{j\in J} Shares_{i,j,t}}. \quad (6)$$

Interpretation. These are ownership-weighted averages. A fund that owns more shares of firm i receives more weight in describing firm i's shareholder base.

2.7 Mechanism variables

The mechanism tests use several additional data sources:

- **SEC shareholder reports.** N-CSR and N-CSRS filings are searched for ESG-related words. Mentions are manually screened to keep discussions tied to investment decisions or performance.

- **Earnings surprises.** The paper uses seasonally adjusted earnings growth and deviation from analyst forecasts as short-term news signals.
- **RepRisk incidents.** The paper uses severe environmental and social incidents as negative ESG-related news.
- **Flow-performance sensitivity.** Mutual fund FPS is estimated from monthly fund flows and past returns using a rolling window.
- **SEC 2004 disclosure shock.** Funds forced to move from semiannual to quarterly holdings disclosure are compared with funds that already voluntarily disclosed quarterly.
- **Morningstar sustainability ratings.** The March 2016 release of Morningstar globes is used to study whether end investors of long-term funds have stronger ESG preferences.
- **Climate Change News Index.** The Engle et al. climate-news index measures time-varying climate attention.

3 Models and empirical specifications (all equations + interpretation)

3.1 Investor-level portfolio ESG regression

The first core regression is run at the investor-quarter level:

$$InvestorESG_{j,t+1} = \alpha_t + \beta_1 Horizon_{j,t} + \beta_2 X_{j,t} + \varepsilon_{j,t}. \tag{7}$$

Here, $InvestorESG_{j,t+1}$ is the value-weighted average ESG score of investor j's portfolio. $Horizon_{j,t}$ is measured by turnover, churn, or adjusted churn. The controls include portfolio size, number of portfolio stocks, and weighted-average characteristics of portfolio firms such as market capitalization, book-to-market ratio, and past returns. For mutual funds, the specification also includes investment-objective-by-quarter fixed effects.

Interpretation. Because turnover and churn are *short-horizon* measures, the prediction is $\beta_1 < 0$. A negative coefficient means higher trading frequency is associated with lower portfolio ESG scores, or equivalently, longer-horizon investors tilt toward higher-ESG stocks.

3.2 ESG-discussion regression

To test whether long-horizon funds explicitly discuss ESG considerations, the paper estimates

$$D(ESGMention)_{j,t} = \alpha_t + \beta_1 Horizon_{j,t} + \beta_2 X_{j,t} + \varepsilon_{j,t}. \tag{8}$$

The dependent variable equals one when a shareholder report contains an ESG-related discussion in the context of investment decisions or performance.

Interpretation. Again, the expected sign is negative. Lower turnover and lower churn should predict a higher probability of mentioning ESG.

3.3 Firm-level shareholder-horizon regression

The second core regression is run at the firm-year level:

$$ShareholderHorizon_{i,t+1} = \alpha_t + \beta_1 ESG_{i,t} + \beta_2 Z_{i,t} + \varepsilon_{i,t}. \tag{9}$$

The dependent variable is a shareholder-base horizon proxy: mutual fund turnover, mutual fund churn, 13f churn, or transient ownership divided by total 13f ownership. Controls include firm size, book-to-market, dividend yield, profitability, past return, return volatility, stock turnover, industry fixed effects, and year fixed effects.

Interpretation. Because the dependent variables are mostly short-horizon measures, the prediction is $\beta_1 < 0$. A high-ESG firm should have less shareholder turnover, less shareholder churn, or a lower share of transient institutional ownership.

3.4 Rating-disagreement and measurement-error tests

The paper first repeats the firm-level regression using Refinitiv and Sustainalytics ESG scores. It then uses an IV approach in which MSCI ESG is instrumented with the other two ESG ratings:

$$MSCI\ ESG_{i,t} = \pi_0 + \pi_1 Sustainalytics_{i,t} + \pi_2 Refinitiv_{i,t} + \Pi Z_{i,t} + u_{i,t}, \tag{10}$$

$$ShareholderHorizon_{i,t+1} = \alpha_t + \beta_1 \widehat{MSCI\ ESG}_{i,t} + \beta_2 Z_{i,t} + \varepsilon_{i,t}. \tag{11}$$

Interpretation. If rating noise is largely idiosyncratic across providers, using one rating to instrument another helps recover the common ESG component. The paper's result survives this test.

3.5 Trading response to earnings shortfalls

To test whether long-horizon investors respond differently to short-term news, the paper estimates:

$$\begin{aligned} Trade_{i,j,t} = \alpha_{j,t} &+ \beta_1 EarningsShortfall_{i,t} + \beta_2 \left(LongTermFund_{j,t} \times EarningsShortfall_{i,t}\right) \\ &+ \Gamma Controls_{i,t} + \varepsilon_{i,j,t}. \end{aligned} \tag{12}$$

The dependent variable is alternatively an indicator for selling, liquidating, or buying the stock.

Interpretation. If long-term investors are patient with short-term earnings misses, the interaction coefficient should be negative for selling and liquidation, and positive for buying. That is exactly the pattern in the paper.

3.6 Trading response to environmental and social incidents

The analogous specification for negative ES incidents is:

$$\begin{aligned} Trade_{i,j,t} = \alpha_{j,t} &+ \beta_1 ESIncident_{i,t-1} + \beta_2 \left(LongTermFund_{j,t} \times ESIncident_{i,t-1}\right) \\ &+ \Gamma Controls_{i,t} + \varepsilon_{i,j,t}. \end{aligned} \tag{13}$$

Some specifications further include stock-by-quarter fixed effects, so the paper compares long- and short-horizon funds holding the same stock at the same time.

Interpretation. If long-term investors pay more attention to ESG-related information, β_2 should be positive for selling and liquidation, and negative for buying. The paper finds this pattern.

3.7 Flow-performance sensitivity and limits to arbitrage

The paper estimates flow-performance sensitivity, then relates it to investor horizon and portfolio ESG:

$$FPS_{j,t} = \alpha_t + \beta_1 Horizon_{j,t} + \beta_2 X_{j,t} + \varepsilon_{j,t}, \tag{14}$$

$$FundESG_{j,t+1} = \alpha_t + \beta_1 FPS_{j,t} + \beta_2 X_{j,t} + \varepsilon_{j,t}. \tag{15}$$

Interpretation. Short-horizon funds have higher FPS, and higher FPS predicts lower portfolio ESG. This is consistent with a limits-to-arbitrage story: managers who face more flighty flows cannot easily hold positions whose payoff is long horizon or slow to be recognized by the market.

3.8 SEC disclosure shock and IV design

The 2004 SEC rule increased mandatory portfolio-disclosure frequency. The reduced form is:

$$FundESG_{j,t+1} = \alpha_j + \gamma_t + \beta \left(TreatedFund_j \times PostDisclosure_t\right) + \Gamma X_{j,t} + \varepsilon_{j,t}. \tag{16}$$

The 2SLS version uses the rule-induced change as an instrument for horizon:

$$Horizon_{j,t} = \alpha_j + \gamma_t + \beta \left(TreatedFund_j \times PostDisclosure_t\right) + \Gamma X_{j,t} + \eta_{j,t}, \tag{17}$$

$$FundESG_{j,t+1} = \alpha_j + \gamma_t + \beta_1 \widehat{Horizon}_{j,t} + \Gamma X_{j,t} + \varepsilon_{j,t}. \tag{18}$$

Interpretation. Treated funds are those forced to disclose more frequently after the SEC rule. If this shortens their horizons and lowers their ESG tilt, that supports a causal interpretation from horizon to ESG preference.

3.9 Clientele-catering tests

The Morningstar sustainability-rating test uses fund flows:

$$\begin{aligned} Flows_{j,t} = \alpha_j + \alpha_t + \beta_1 \left(LongTermFund_{j,t} \times Globe_j \times Post_t\right) \\ + \beta_2 \left(LongTermFund_{j,t} \times Globe_j\right) + \beta_3 \left(LongTermFund_{j,t} \times Post_t\right) + \Gamma X_{j,t} + \varepsilon_{j,t}. \end{aligned} \tag{19}$$

The climate-news test estimates:

$$FundESG_{j,t+1} = \alpha_t + \beta_1 Horizon_{j,t} + \beta_2 \left(Horizon_{j,t} \times CCNI_t\right) + \Gamma X_{j,t} + \varepsilon_{j,t}. \tag{20}$$

Interpretation. If long-term funds tilt toward ESG mainly to cater to a more ESG-sensitive clientele, long-term funds should receive stronger flow responses to Morningstar sustainability ratings and should increase ESG tilts more when climate news is salient. The paper does not find these patterns.

4 Identification and mechanisms

4.1 What identifies the baseline relation

The investor-level design compares funds or institutions with different horizons while controlling for portfolio characteristics and, for mutual funds, investment-objective-by-quarter fixed effects. This helps compare funds with similar stated investment styles in the same quarter.

The firm-level design asks whether high-ESG firms attract longer-horizon shareholders after controlling for firm characteristics, industry fixed effects, and year fixed effects.

Interpretation. The baseline results are not a randomized experiment. They establish a robust cross-sectional relation. The mechanism tests then try to distinguish whether the relation reflects information processing, arbitrage constraints, or clientele catering.

4.2 Why the result is not simply an ESG-fund mandate

The mutual fund sample excludes explicit ESG and SRI funds. This matters because ESG funds might mechanically hold high-ESG firms and might also have lower turnover because their investment universe is more constrained.

Interpretation. The result is stronger because it appears among ordinary active domestic equity funds, not just among funds that advertise ESG investing.

4.3 Why rating disagreement is not fatal

ESG ratings from MSCI, Sustainalytics, and Refinitiv differ. The paper addresses this in two ways:

- repeat the firm-level tests using the alternative rating providers;
- use the alternative ratings as instruments for MSCI ESG, treating disagreement as measurement error.

Interpretation. If three noisy measures point in the same direction, the common component is likely related to firms' underlying ESG profiles rather than a provider-specific scoring artifact.

4.4 Information channel

The information-channel hypothesis is not simply that long-term funds trade less. Long-term funds generally trade less, but the key prediction is more subtle:

- they should be patient with short-term earnings disappointments;
- they should react strongly to negative ESG-related news if they view that information as relevant to long-run value or long-run risk.

The paper finds exactly this contrast. Long-term funds are less likely to sell after negative earnings surprises but more likely to sell or liquidate after ES incidents.

Interpretation. The evidence supports a specialization story. Short-horizon investors respond more to high-frequency earnings news. Long-horizon investors pay more attention to longer-horizon ESG-related information.

4.5 Limits-to-arbitrage channel

The limits-to-arbitrage channel says that even if high-ESG stocks are attractive in the long run, some managers cannot easily exploit that opportunity. Open-end mutual funds may face outflows after short-term underperformance. If the payoff to ESG investing is slow, managers with high flow-performance sensitivity may avoid ESG tilts.

The paper provides two pieces of evidence:

- funds with higher turnover or adjusted churn have higher flow-performance sensitivity;
- funds with higher flow-performance sensitivity hold lower-ESG portfolios.

The SEC 2004 disclosure rule provides a stronger test. Funds forced to disclose more frequently shortened their horizons and moved away from high-ESG portfolios.

Interpretation. This is the paper's most causal-looking evidence. The shock changes the investment environment faced by funds, rather than merely comparing funds that already differ in horizon.

4.6 Clientele-catering channel

The catering channel requires a specific condition: investors in long-term funds must have stronger ESG preferences than investors in short-term funds. The paper finds little support for that condition.

Morningstar sustainability ratings generate flows: low-rated funds lose money and high-rated funds gain money. But this flow response does not differ meaningfully between long- and short-term funds.

Climate-news salience also fails to support catering. During periods of greater climate-news intensity, long-term funds do not increase their ESG tilt more than short-term funds.

Interpretation. The paper does not prove catering is irrelevant in all settings. It shows that the two empirical signatures one would expect under clientele catering are not present in this sample.

4.7 Remaining limitations

The paper is careful about mechanisms, but several limitations remain:

- ESG scores are imperfect proxies for true ESG quality.
- Horizon measures are behavioral proxies and may capture other aspects of fund strategy.
- The baseline relation is observational, although the SEC shock strengthens the causal interpretation.
- The paper studies revealed portfolio tilts, not the welfare consequences of ESG investing.

Interpretation. The strongest interpretation is that investor horizon is an important organizing variable for understanding ESG portfolio allocation and related trading behavior.

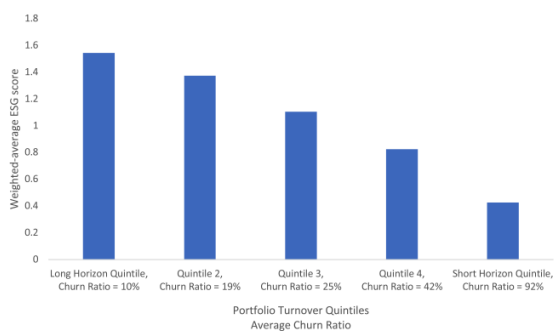
5 Tables and figures

5.1 Figure 1: portfolio ESG scores by investor horizon

Read: Figure 1 sorts investors by turnover or churn. For mutual funds, the long-horizon quintile has a much higher average portfolio ESG score than the short-horizon quintile. The same monotonic pattern appears for 13f institutions.

Economic message. The raw data already show the paper's central pattern: longer-horizon investors hold higher-ESG portfolios. The rest of the paper asks whether this survives controls and what mechanism explains it.

Panel A. Mutual Funds Sorted by Turnover Ratios



Panel B. 13F Institutions Sorted by Churn Ratios

Figure 1. Portfolio holdings' ESG scores and investor horizons. This figure shows the average ESG scores of investors' portfolio holdings in relation to their horizons. Investor horizons are measured by the portfolio turnover ratio for mutual funds and the portfolio churn ratios for 13f institutions. Investors with the lowest portfolio turnover ratios (churn ratios) are defined as having a long horizon. Panel A corresponds to mutual funds and Panel B to 13f institutions. Each mutual fund's (13f institution's) ESG score is calculated as the value-weighted average ESG score of its portfolio companies. (Color figure can be viewed at wileyonlinelibrary.com)

Table I
Summary Statistics

Panel A reports summary statistics for the full sample of fund-quarters used in the mutual fund-level analyses. Panel B reports the summary statistics for 13f institutions. Panel C reports summary statistics for firm-years used in the firm-level analyses. The sample period is 2000 to 2018.

Panel A: Mutual Fund-Level Variables					
$N = 93{,}290$	Mean	Stdev	P25	Median	P75
Portfolio Turnover (Annual)	0.71	0.49	0.31	0.57	0.98
Churn Ratio (Quarterly)	0.37	0.23	0.20	0.32	0.48
Adjusted Churn Ratio (Quarterly)	0.14	0.10	0.06	0.11	0.18
ln(Fund TNA)	5.78	1.90	4.43	5.79	7.11
Weighted-Average Values for Portfolio Stocks:					
MSCI ESG Score	1.06	1.85	−0.37	0.63	2.40
ln(Market Cap)	9.87	1.65	8.40	10.59	11.28
Book-to-Market	0.50	0.20	0.36	0.47	0.59
12-Month Return	0.19	0.23	0.06	0.18	0.30
Panel B: 13f Institution-Level Variables					
$N = 166{,}185$	Mean	Stdev	P25	Median	P75
Churn Ratio (Quarterly)	0.35	0.32	0.13	0.24	0.45
Adjusted Churn Ratio (Quarterly)	0.12	0.12	0.04	0.08	0.16
ln(Total Holdings value)	19.73	1.82	18.47	19.43	20.82
Weighted-Average Values for Portfolio Stocks:					
MSCI ESG Score	1.53	1.88	0.07	1.24	3.06
ln(Market Cap)	10.78	1.16	10.28	11.16	11.59
Book-to-Market	0.52	0.30	0.37	0.45	0.58
12-Month Return	0.18	0.22	0.06	0.16	0.27
Panel C: Firm-Level Variables					
$N = 26{,}217$	Mean	Stdev	P25	Median	P75
MSCI ESG Score	−0.03	2.32	−1	0	1
Sustainalytics ESG score	55.75	8.67	49	55	62
Refinitiv ESG score	39.70	18.78	25.09	36.54	52.09
ln(Market Cap)	7.26	1.59	6.06	7.05	8.23
Book-to-Market	0.61	0.47	0.29	0.50	0.78
12-Month Return (%)	14.57	47.25	−13.04	9.84	34.11
Return Volatility (%)	10.77	6.22	6.40	9.17	13.32
Dividend Yield (%)	1.28	2.01	0.00	0.15	2.00
Profitability	0.092	0.142	0.036	0.105	0.162
Daily Turnover (%)	0.24	0.52	0.07	0.13	0.22
Mutual Fund Ownership(%)	14.21	8.53	7.64	12.68	20.01
13f Institution Ownership(%)	69.60	22.04	56.67	74.79	87.06
Weighted-Average Values for the Firms' Shareholders:					
Mutual Fund Shareholder Turnover (%)	42.98	13.99	31.77	39.84	51.87
Mutual Fund Shareholder Churn Ratio (%)	26.77	8.87	20.24	25.35	31.95
13f Institution Shareholder Churn Ratio (%)	23.56	6.74	18.52	22.76	27.56
Transient/Total 13f Institution Ownership (%)	17.65	14.68	4.70	16.27	27.25

5.2 Table I: summary statistics

Read: Table I reports investor-level and firm-level summary statistics. Mutual funds have average annual turnover of about 0.71, quarterly churn of about 0.37, and adjusted churn of about 0.14. 13f institutions have average quarterly churn of about 0.35 and adjusted churn of about 0.12. Firm-level average MSCI ESG is close to zero, while portfolio-weighted ESG scores are higher because large firms tend to have higher ESG scores.

Economic message. The table shows meaningful variation in both ESG scores and investor horizons. This is necessary for the cross-sectional tests.

5.3 Table II: investor-level ESG tilt

Read: Table II regresses investor-level portfolio ESG on horizon proxies. In the mutual fund sample, turnover, churn, and adjusted churn all have negative and significant coefficients. In the 13f sample, churn and adjusted churn are also negative and significant.

Economic message. Since higher turnover and churn mean shorter horizons, Table II says that long-horizon investors hold higher-ESG portfolios. This is the central investor-level result.

Table II
Investor-Level Analysis: Portfolio ESG Score and Investor Horizon

This table reports results from an investor-level regression of the average ESG score for portfolio holdings on investor horizon: $InvestorESG_{j,t} = \alpha + \beta_1 Horizon_{j,t} + \beta_2 X_{j,t} + \epsilon_{j,t}$. *InvestorESG* is the weighted-average ESG score of an investor's portfolio holdings. Investor *Horizon* for mutual funds is proxied by turnover ratio, estimated fund churn ratio, or adjusted churn ratio. The *Horizon* for 13f investors is proxied by churn ratios and adjusted churn ratios. Fund and 13f churn ratios are estimated over the previous four quarters. The fractional 12-month return rank for funds in the bottom within-objective quintile (*Low*) is defined as $Min(Rank_{t-1}, 0.2)$. Funds in the three medium return quintiles (*Mid*) are grouped together and given ranks accordingly to $Min(0.6, Rank_{t-1} - Low)$. The rank for the top return quintile (*High*) is defined as $Rank_{t-1} - Mid - Low$. All specifications include investment objective-by-quarter fixed effects or quarter fixed effects. Standard errors, shown in parentheses, are two-way clustered by fund and by quarter. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period covers 2000 to 2018.

Dependent Variable: Investor-Level ESG Score

Sample	Mutual Funds			13f Institutions	
	(1)	(2)	(3)	(4)	(5)
Fund Turnover Ratio	−0.0968***				
	(0.0197)				
Fund Churn Ratio		−0.216***		−0.321***	
		(0.0402)		(0.0310)	
Adjusted Churn Ratio			−0.555***		−0.820***
			(0.0962)		(0.0755)
Portfolio Average Ln(Market Cap)	0.647***	0.644***	0.644***	0.659***	0.653***
	(0.0480)	(0.0479)	(0.0479)	(0.0463)	(0.0461)
Portfolio Average Book-to-Market Ratio	−0.0380	−0.0301	−0.0359	0.0431	0.0404
	(0.167)	(0.166)	(0.167)	(0.0664)	(0.0663)
Portfolio Average 12-Month Return	−0.0238	−0.0176	−0.00221	−0.393*	−0.359*
	(0.0634)	(0.0641)	(0.0629)	(0.199)	(0.189)
Number of Portfolio Stocks (in 100s)	−0.0413***	−0.0448***	−0.0427***	−0.0135***	−0.0138***
	(0.00807)	(0.00807)	(0.00797)	(0.00282)	(0.00280)
Fund Return Rank – Low	−0.0828	−0.0877	−0.0971		
	(0.164)	(0.164)	(0.165)		
Fund Return Rank – Mid	−0.0887	−0.0835	−0.0895		
	(0.0643)	(0.0644)	(0.0644)		
Fund Return Rank – High	−1.110***	−1.099***	−1.143***		
	(0.294)	(0.294)	(0.294)		
Ln(Fund TNA)	−0.00589	−0.00604	−0.00568		
	(0.00600)	(0.00602)	(0.00592)		
Ln(Total Holdings Value) for 13f				0.00133	0.00429
				(0.00729)	(0.00725)
Observations	93,290	93,290	93,290	166,185	172,277
Adjusted R^2	0.835	0.835	0.835	0.722	0.719
Quarter FE	NA	NA	NA	Y	Y
Investment Objective-by-Quarter FE	Y	Y	Y	N	N

Table III
Mutual Fund Investment Horizon and ESG Discussions in Their Shareholder Reports

This table examines the association between a mutual fund's investment horizon and the probability that the fund mentions ESG-related words in their semiannual shareholder reports. The dependent variable is an indicator that denotes at least one ESG-related keyword is mentioned in its shareholder report in the context of investment decisions or performance during the reporting period. The investment horizon of a mutual fund is proxied by its portfolio turnover ratio, churn ratio, or adjusted churn ratio, all of which are estimated over the previous four quarters. The fractional 12-month return rank for funds in the bottom within-objective quintile (*Low*) is defined as $Min(Rank_{t-1}, 0.2)$. Funds in the three medium return quintiles (*Mid*) are grouped together and given ranks according to $Min(0.6, Rank_{t-1} - Low)$. The rank for the top return quintile (*High*) is defined as $Rank_{t-1} - Mid - Low$. A higher ratio indicates a shorter investment horizon. Robust standard errors are shown in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level respectively. The sample period is 2000 to 2018.

Dependent Variable: *D(ESG Mention)*∗100

	(1)	(2)	(3)
Fund Turnover Ratio	−0.889***		
	(0.263)		
Fund Churn Ratio		−1.841***	
		(0.574)	
Adjusted Churn Ratio			−2.531**
			(1.283)
Holdings Ln(Market Cap)	−0.430***	−0.436***	−0.430***
	(0.0876)	(0.0877)	(0.0879)
Holdings Book-to-Market	−3.837***	−3.798***	−3.796***
	(0.765)	(0.763)	(0.764)
Fund Return Rank – Low	4.298	4.229	4.477
	(3.609)	(3.606)	(3.609)
Fund Return Rank – Mid	−0.492	−0.467	−0.475
	(0.813)	(0.813)	(0.813)
Fund Return Rank – High	−2.427	−2.298	−2.720
	(4.437)	(4.439)	(4.438)
Ln(Fund TNA)	0.131*	0.134*	0.157**
	(0.0729)	(0.0724)	(0.0718)
Number of Portfolio Stocks (in 100s)	−0.783***	−0.818***	−0.823***
	(0.107)	(0.106)	(0.105)
Observations	32,926	32,926	32,926
Adjusted R^2	0.030	0.030	0.030
Time FE	Y	Y	Y

5.4 Table III: ESG mentions in shareholder reports

Read: Table III studies whether mutual funds mention ESG-related considerations in shareholder reports. The dependent variable is an ESG-mention indicator multiplied by 100. Turnover, churn, and adjusted churn have negative coefficients.

Economic message. Longer-horizon funds do not merely happen to own higher-ESG stocks. They are also more likely to discuss ESG considerations in their own communications. This supports the interpretation that ESG is part of their investment process.

5.5 Table IV: firm-level shareholder horizon

Read: Table IV reverses the perspective. Instead of asking whether long-term investors hold high-ESG firms, it asks whether high-ESG firms have longer-term shareholders. The MSCI ESG coefficient is negative and significant for mutual fund turnover, mutual fund churn, 13f churn, and transient ownership.

Economic message. High-ESG firms have a more patient shareholder base. The investor-level and firm-level evidence reinforce each other.

Table IV
Firm-Level Analysis: ESG Score and Shareholders' Horizons

This table reports results of regressions of the investor horizon of shareholders on firms' MSCI ESG scores and control variables. Observations are at the firm-year level. In column (1), the dependent variable is the weighted-average turnover ratio of a firm's mutual fund shareholders. In column (2), the dependent variable is the weighted-average churn ratio of a firm's mutual fund shareholders. In column (3), the dependent variable is the weighted-average churn ratio of a firm's 13f institution shareholders. In column (4), the dependent variable is the ratio between transient (*TRA*) institutional ownership and total 13f institution ownership in the firm. Year fixed effects and industry fixed effects are included across specifications. Standard errors, shown in parentheses, are double clustered at the stock and year level. *, **, and *** indicate significance at the 10%, 5%, and 1% level respectively. The sample period is 2000 to 2018.

Shareholder Horizon Proxied by:	MF Turnover (1)	MF *Churn* (2)	13f *Churn* (3)	$TRA_{t+1}/13fOwn$ (4)
MSCI ESG Score	−0.222**	−0.145***	−0.0984***	−0.0964**
	(0.102)	(0.0461)	(0.0319)	(0.0423)
Ln(Market Cap)	0.555*	−0.571***	−0.593***	−0.683***
	(0.288)	(0.161)	(0.0714)	(0.124)
Book-to-Market Ratio	−2.252***	−0.896***	−0.0996	−0.785***
	(0.334)	(0.228)	(0.187)	(0.266)
Dividend Yield	−0.814***	−0.364***	−0.367***	−0.248***
	(0.125)	(0.0486)	(0.0477)	(0.0807)
Profitability Ratio	−0.657	0.0692	−2.037***	−1.052
	(1.397)	(0.782)	(0.566)	(0.749)
Past 12-Month Return	0.0608***	0.0289***	0.0115***	0.0207***
	(0.00601)	(0.00295)	(0.00131)	(0.00343)
Return Volatility	0.265***	0.198***	0.200***	0.192***
	(0.0460)	(0.0251)	(0.0282)	(0.0301)
Stock Turnover	1.186***	0.709***	1.102***	0.574*
	(0.308)	(0.221)	(0.154)	(0.327)
Underlying Fund Flow Volatility	1.474***	1.839***		
	(0.251)	(0.253)		
Observations	26,207	26,217	26,217	26,217
Adjusted R^2	0.451	0.442	0.500	0.684
Industry (SIC2) Fixed-Effects	Y	Y	Y	Y
Year Fixed-Effects	Y	Y	Y	Y

Table V
Firm-Level Analysis: Refinitiv and Sustainalytics ESG Scores

This table reports results of regressions of investor horizon on firms' ESG profiles, as measured by both Refinitiv ESG scores and Sustainalytics scores. For the sample using Refinitiv ESG scores, the sample period is 2002 to 2018. For the sample using Sustainalytics ESG scores, the period is 2009 to 2017. Observations are at the firm-year level. Year fixed effects and industry fixed effects are included across specifications. Standard errors, shown in parentheses, are double clustered at the stock and year level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable(%)	MF Turnover (1)	MF *Churn* (2)	13f *Churn* (3)	$TRA_{t+1}/13fOwn$ (4)	MF Turnover (5)	MF *Churn* (6)	13f *Churn* (7)	$TRA_{t+1}/13fOwn$ (8)
Refinitiv ESG Score	−0.0227**	−0.0123**	−0.0167***	−0.0137**				
	(0.0102)	(0.00521)	(0.00481)	(0.00572)				
Sustainalytics ESG Score					−0.0363*	−0.0226**	−0.0252**	−0.0152
					(0.0196)	(0.00982)	(0.0103)	(0.0129)
Ln(Market Cap)	−0.485***	−1.243***	−0.799***	−0.677***	−0.462**	−1.104***	−1.173***	−1.117***
	(0.145)	(0.0778)	(0.0678)	(0.0721)	(0.182)	(0.0954)	(0.102)	(0.110)
Book-to-Market Ratio	−1.615***	−0.557**	−0.211	−0.162	−0.769	0.139	0.139	−0.258
	(0.449)	(0.220)	(0.237)	(0.260)	(0.546)	(0.286)	(0.315)	(0.429)
Dividend Yield	−0.949***	−0.465***	−0.388***	−0.254***	−0.850***	−0.447***	−0.459***	−0.320***
	(0.0858)	(0.0464)	(0.0399)	(0.0433)	(0.101)	(0.0532)	(0.0545)	(0.0648)
Profitability Ratio	0.215	1.184	−2.288***	1.446**	−5.647***	−1.295	−1.741*	1.244
	(1.178)	(0.752)	(0.631)	(0.629)	(1.764)	(0.936)	(1.007)	(1.284)
Past 12-Month Return	0.0381***	0.0238***	0.0121***	0.0192***	0.0258***	0.0155***	0.00627***	0.0153***
	(0.00293)	(0.00157)	(0.00134)	(0.00187)	(0.00384)	(0.00227)	(0.00232)	(0.00313)
Return Volatility	0.0488	0.0235	0.0938***	−0.0183	0.169***	0.0926***	0.151***	0.0725**
	(0.0398)	(0.0204)	(0.0188)	(0.0229)	(0.0586)	(0.0270)	(0.0250)	(0.0364)
Stock Turnover	9.456***	4.306***	7.003***	6.093***	1.683*	0.919**	1.848***	1.040
	(2.011)	(0.805)	(0.892)	(0.984)	(0.929)	(0.367)	(0.419)	(0.656)
Underlying Fund Flow Volatility	2.608***	2.114***			2.705***	2.436***		
	(0.259)	(0.140)			(0.349)	(0.252)		
Observations	11,653	11,653	11,653	11,653	5,897	5,898	5,898	5898
Adjusted R^2	0.553	0.562	0.539	0.846	0.425	0.532	0.493	0.778
Industry (SIC2) Fixed-Effects	Y	Y	Y	Y	Y	Y	Y	Y
Year Fixed-Effects	Y	Y	Y	Y	Y	Y	Y	Y

Table VI
Firm-Level Analysis: Errors-in-Variable Approach

This table reports results of regressions of investor horizon on firms' ESG profiles, as measured by both MSCI ESG scores and the errors-in-variable approach of Berg et al. (2024). The sample is constrained to the period of 2008 to 2017 for the set of firms with available ESG rating in MSCI, Sustainalytics, and Refinitiv. Observations are at the firm-year level. In columns (1) to (4), we use the raw MSCI ESG score for each firm. In columns (5) to (8), we instrument for a firm-year's MSCI ESG score using its Sustainalytics and Refinitiv ESG scores. Year fixed effects and industry fixed effects are included across specifications. Standard errors, shown in parentheses, are double clustered at the stock and year level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable(%)	MF Turnover (1)	MF *Churn* (2)	13f *Churn* (3)	$TRA_{t+1}/13fOwn$ (4)	MF Turnover (5)	MF *Churn* (6)	13f *Churn* (7)	$TRA_{t+1}/13fOwn$ (8)
MSCI ESG Score	−0.220***	−0.144***	−0.109***	−0.158***				
	(0.0612)	(0.0295)	(0.0281)	(0.0392)				
MSCI ESG Score (instrumented)					−0.384**	−0.257***	−0.201**	−0.141
					(0.151)	(0.0773)	(0.0818)	(0.102)
Ln(Market Cap)	−0.286	−1.006***	−1.072***	−0.841***	−0.117	−0.889***	−0.976***	−0.859***
	(0.190)	(0.0950)	(0.101)	(0.103)	(0.237)	(0.124)	(0.137)	(0.142)
Book-to-Market Ratio	−0.317	0.161	0.113	0.0951	−0.235	0.218	0.160	0.0863
	(0.600)	(0.279)	(0.361)	(0.465)	(0.601)	(0.273)	(0.364)	(0.473)
Dividend Yield	−0.806***	−0.448***	−0.459***	−0.272***	−0.789***	−0.436***	−0.449***	−0.274***
	(0.107)	(0.0573)	(0.0586)	(0.0694)	(0.107)	(0.0572)	(0.0575)	(0.0691)
Profitability Ratio	−5.454***	−0.996	−2.053*	1.483	−5.520***	−1.042	−2.091**	1.490
	(1.901)	(0.970)	(1.072)	(1.379)	(1.885)	(0.961)	(1.058)	(1.364)
Past 12-Month Return	0.0315***	0.0183***	0.00884***	0.0184***	0.0319***	0.0186***	0.00906***	0.0183***
	(0.00447)	(0.00217)	(0.00211)	(0.00324)	(0.00444)	(0.00216)	(0.00209)	(0.00324)
Return Volatility	0.0259	0.0527*	0.0801***	−0.0146	0.0233	0.0508*	0.0786***	−0.0143
	(0.0546)	(0.0287)	(0.0279)	(0.0401)	(0.0543)	(0.0284)	(0.0276)	(0.0399)
Stock Turnover	10.20***	3.698***	6.963***	6.336***	10.29***	3.766***	7.018***	6.326***
	(2.368)	(1.008)	(0.939)	(1.527)	(2.381)	(1.023)	(0.939)	(1.513)
Underlying Fund Flow Volatility	2.900***	2.244***			2.892***	2.238***		
	(0.370)	(0.190)			(0.367)	(0.188)		
Observations	5,052	5,052	5,052	5,052	5,053	5,053	5,053	5053
Adjusted R^2	0.478	0.602	0.563	0.811	0.136	0.265	0.251	0.073
Industry (SIC2) Fixed-Effects	Y	Y	Y	Y	Y	Y	Y	Y
Year Fixed-Effects	Y	Y	Y	Y	Y	Y	Y	Y

5.6 Tables V and VI: alternative ESG ratings and measurement error

Read: Table V repeats the firm-level tests using Refinitiv and Sustainalytics scores. Table VI treats rating disagreement as an errors-in-variables problem and instruments MSCI ESG with Sustainalytics and Refinitiv.

Economic message. The horizon-ESG relation survives rating-provider disagreement. This is important because ESG ratings are known to vary across providers.

5.7 Table VII: trading after negative earnings surprises

Read: Table VII studies trading after negative earnings surprises. Non-long-term funds are more likely to sell or liquidate after earnings disappointments. The interaction with the long-term-fund indicator is negative for selling and liquidation, and positive for buying.

Economic message. Long-term investors are more patient with short-run earnings news. This supports the idea that their horizon changes how they process information.

5.8 Table VIII: trading after negative ES incidents

Read: Table VIII studies trading after negative environmental or social incidents. Long-term funds are significantly more likely to sell or liquidate exposed stocks, and less likely to buy them. The estimates remain similar when stock-by-time fixed effects are added.

Economic message. Long-term investors are not simply inactive. They are patient about earnings

Table VII

Mutual Fund Trading in Response to Negative Earnings Surprises

This table reports results of regressions of the interquarter trading decisions of mutual funds in response to recent negative earnings surprises of portfolio stocks. The sample only includes positions in which the fund's holding in the previous quarter was positive. The outcome variable *D(sell)* indicates that a fund reduces its position in a given stock between quarters $t-1$ and t. The outcome variable *D(liq)* indicates that a fund liquidates its position in a given stock between quarters $t-1$ and t so that it holds zero shares at t. The outcome variable *D(buy)* indicates that a fund increases its position in a given stock between quarters $t-1$ and t. *D(Neg. Earnings Surprise)* is an indicator for a negative earnings surprise for a stock-quarter. *D(Long-term Fund)* is an indicator for a mutual fund whose four-quarter trailing-average churn ratio falls below the 30th percentile in the cross section. Standard errors, shown in parentheses, are double-clustered at the fund and quarter level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Measurement of Earnings Surprise	Seasonal-Adjusted Earnings Growth			Deviation from Analyst Forecast		
Dependent Variable (scaled by *100)	*D(sell)*	*D(liq)*	*D(buy)*	*D(sell)*	*D(liq)*	*D(buy)*
	(1)	(2)	(3)	(4)	(5)	(6)
D(Neg. Earnings Surprise)	1.208***	2.358***	−0.520***	1.231***	2.400***	−0.527***
	(0.133)	(0.194)	(0.123)	(0.141)	(0.207)	(0.125)
*D(Neg. Earnings Surprise)*D(Long-term Fund)*	−1.871***	−2.234***	1.680***	−2.020***	−2.340***	1.782***
	(0.161)	(0.182)	(0.153)	(0.173)	(0.184)	(0.158)
Past 12-Month return	−0.0573	−4.909***	−0.602***	−0.0611	−4.907***	−0.598***
	(0.242)	(1.293)	(0.197)	(0.241)	(1.292)	(0.198)
Return Volatility	17.24***	37.76***	11.98***	17.22***	37.66***	11.97***
	(3.666)	(6.792)	(2.742)	(3.663)	(6.804)	(2.747)
Stock Turnover	0.105***	0.0891***	−0.0768***	0.106***	0.0897***	−0.0767***
	(0.0321)	(0.0175)	(0.0143)	(0.0322)	(0.0175)	(0.0142)
Ln(Market Cap)	0.268**	−2.407***	1.652***	0.269**	−2.403***	1.652***
	(0.112)	(0.105)	(0.115)	(0.111)	(0.105)	(0.115)
Book-to-Market Ratio	−0.346	0.495**	0.258	−0.349	0.481*	0.257
	(0.213)	(0.243)	(0.154)	(0.215)	(0.244)	(0.155)
Dividend Yield	2.677***	4.199***	−1.063	2.685***	4.188***	−1.076
	(0.921)	(1.363)	(0.958)	(0.925)	(1.359)	(0.958)
Observations	4,877,697	4,877,697	4,877,697	4,877,697	4,877,697	4,877,697
Adjusted R^2	0.267	0.157	0.245	0.267	0.157	0.245
Fund-by-Time FE	Y	Y	Y	Y	Y	Y

Table VIII

Mutual Fund Trading in Response to ES Incidents

This table reports results of regressions of the interquarter trading decisions of mutual funds in response to recent negative environmental- or social-related (ES) incidents. The sample only includes positions in which the fund's holding in the previous quarter was positive. The outcome variable *D(sell)* indicates that a fund reduces its position in a given stock between quarters $t-1$ and t. The outcome variable *D(liq)* indicates that a fund liquidates its position in a given stock between quarters $t-1$ and t so that it holds zero shares at t. The outcome variable *D(buy)* indicates that a fund increases its position in a given stock between quarters $t-1$ and t. *D(ES Incident)* is an indicator for a reported RepRisk incident in the previous quarter. *D(Long-term Fund)* is an indicator for a mutual fund whose four-quarter trailing-average churn ratio falls below the 30th percentile in the cross section. Standard errors, shown in parentheses, are double-clustered at the fund and quarter level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable (Scaled by *100)	*D(sell)*	*D(liq)*	*D(buy)*	*D(sell)*	*D(liq)*	*D(buy)*
	(1)	(2)	(3)	(4)	(5)	(6)
D(ES Incident)	−1.930	−1.200	0.769			
	(1.564)	(1.347)	(1.146)			
*D(ES Incident)*D(Long-term Fund)*	4.652***	3.244***	−2.825***	4.498***	3.330***	−2.970***
	(1.171)	(0.906)	(0.917)	(1.129)	(0.924)	(0.692)
Past 12-Month Return	−0.174	−5.197***	−0.590***			
	(0.216)	(1.348)	(0.195)			
Return Volatility	17.81***	39.14***	11.92***			
	(3.699)	(6.881)	(2.745)			
Stock Turnover	0.104***	0.0857***	−0.0766***			
	(0.0305)	(0.0185)	(0.0142)			
Ln(Market Cap)	0.264**	−2.424***	1.651***			
	(0.110)	(0.104)	(0.115)			
Book-to-Market Ratio	−0.314	0.580**	0.259			
	(0.213)	(0.248)	(0.156)			
Dividend Yield	2.815***	4.600***	−1.039			
	(0.926)	(1.475)	(0.955)			
Observations	4,877,697	4,877,697	4,877,697	4,877,096	4,877,096	4,877,096
Adjusted R^2	0.267	0.156	0.245	0.286	0.211	0.260
Fund-by-Time FE	Y	Y	Y	Y	Y	Y
Stock-by-Time FE	N	N	N	Y	Y	Y

misses but responsive to ESG-related bad news. This is strong evidence for the information-channel interpretation.

5.9 Table IX: flow-performance sensitivity

Read: Table IX shows that funds with higher turnover or adjusted churn have higher flow-performance sensitivity. It also shows that higher FPS predicts a lower portfolio ESG score.

Economic message. This supports a limits-to-arbitrage channel. Managers who are more exposed to performance-sensitive flows may avoid long-horizon ESG positions because short-run underperformance can trigger outflows.

Table IX

Investor Horizon, Flow-Performance Sensitivity, and Portfolio ESG Score

This table examines the relationship of mutual funds' investor horizon, flow-performance sensitivity (*FPS*), and their *Portfolio ESG*. A fund's *FPS* is estimated by regressing the fund's monthly net flows on its past 12-month average monthly return using a 24-month rolling window. In columns (1) and (2), the dependent variable is *FPS*. In column (3), the dependent variable is the value-average *Portfolio ESG*. The fractional 12-month return rank for funds in the bottom within-objective quintile (*Low*) is defined as $Min(Rank_{t-1}, 0.2)$. Funds in the three medium return quintiles (*Mid*) are grouped together and given ranks according to $Min(0.6, Rank_{t-1} - Low)$. The rank for the top return quintile (*High*) is defined as $Rank_{t-1} - Mid - Low$. Investment objective-by-quarter fixed effects are included across specifications. Standard errors, shown in parentheses, are double-clustered at the fund and quarter level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is 2000 to 2018.

Dependent Variable: Dependent Variable Scaled By:	FPS (*100)	FPS (*100)	Portfolio ESG
	(1)	(2)	(3)
Fund Turnover Ratio	1.125***		
	(0.294)		
Adjusted Churn Ratio		4.282***	
		(0.667)	
Flow-Performance Sensitivity			−0.337***
			(0.0875)
Ln(Fund TNA)	−0.308***	−0.267***	0.00644
	(0.0701)	(0.0725)	(0.00744)
Holdings Ln(Market Cap)	−0.953***	−0.890***	0.645***
	(0.175)	(0.172)	(0.0403)
Holdings Book-to-Market	3.038***	2.989***	0.499*
	(0.846)	(0.852)	(0.286)
Holdings 12-Month Return	−3.258***	−3.740***	0.681**
	(0.544)	(0.552)	(0.297)
Number of Holdings	−0.310**	−0.299**	−0.0843***
	(0.136)	(0.135)	(0.0148)
Return Rank − Low	−8.907***	−8.244***	−0.0275
	(1.934)	(1.920)	(0.208)
Return Rank − Mid	−0.0191	−0.0297	−0.107
	(0.737)	(0.740)	(0.0846)
Return Rank − High	9.909***	9.672***	−1.381***
	(2.839)	(2.839)	(0.396)
Observations	89,582	89,582	89,608
Adjusted R^2	0.092	0.094	0.613
Investment Objective-by-Quarter FE	Y	Y	Y

Table X

SEC 2004 Mandatory Disclosure Requirement as a Shock to Mutual Fund Horizon

This table reports the effect of SEC's 2004 requirement of quarterly portfolio holdings disclosures on mutual fund horizons and their ESG holdings. Column (1) displays the reduced-form regression, where treated funds are defined as those that did not voluntarily disclose holdings quarterly prior to the requirement, and control funds are those that had already been filing quarterly prior to the SEC requirement. Columns (2) through (4) report results from two-stage-least-squared (2SLS) regressions. In the first stage, fund turnover ratios (churn ratios) are regressed on *Treated Funds * Post Disclosure* and control variables. In the second stage, the instrumented turnover ratios (churn ratios) are used as an explanatory variable on fund portfolio ESG scores. All specifications include fund fixed effects and quarter fixed effects. Standard errors, shown in parentheses, are clustered at the fund level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is 2001 to 2008.

	Reduced-Form	2SLS with Instrumental Variable			
Dependent Variable	Fund ESG	Second Stage Fund ESG	First Stage Turnover	Second Stage Fund ESG	First Stage Churn Ratio
	(1)	(2)	(3)	(4)	(5)
*Treated Funds * Post Disclosure*	−0.0430**		0.0630***		0.0178***
	(0.0185)		(0.0125)		(0.00589)
Fund Turnover Ratio		−0.683**			
		(0.333)			
Fund Churn Ratio				−2.414*	
				(1.319)	
Holdings Ln(Market Cap)	0.365***	0.356***	−0.0122**	0.298***	−0.0277***
	(0.0253)	(0.0242)	(0.00564)	(0.0411)	(0.00424)
Holdings Book-to-Market	0.0146	0.111	0.142***	0.216	0.0835***
	(0.113)	(0.109)	(0.0341)	(0.147)	(0.0206)
Holdings 12-Month Return	0.0184	0.0902	0.105***	0.249	0.0955***
	(0.104)	(0.107)	(0.0221)	(0.165)	(0.0110)
Ln(Fund TNA)	−0.0332***	−0.0597***	−0.0388***	−0.0746***	−0.0171***
	(0.00856)	(0.0152)	(0.00334)	(0.0255)	(0.00209)
Number of Portfolio Stocks (in 100s)	−0.169***	−0.116***	0.0772***	−0.100*	0.0285***
	(0.0248)	(0.0408)	(0.00566)	(0.0527)	(0.00258)
Observations	31,385	31,385	31,385	31,385	31,385
R^2	0.749	0.719	0.777	0.633	0.725
Fund FE	Y	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y	Y

5.10 Table X: SEC 2004 disclosure shock

Read: Table X uses the SEC's 2004 increase in mandatory holdings-disclosure frequency. Treated funds are those forced to move from semiannual to quarterly disclosure. The first stage shows that treated funds' turnover and churn rise after the rule. The second stage shows that instrumented turnover and churn predict lower portfolio ESG.

Economic message. When regulation shortens horizons, funds reduce ESG tilts. This is the paper's strongest evidence that horizon itself affects ESG portfolio choice.

5.11 Table XI: Morningstar sustainability ratings

Read: Table XI replicates the idea that Morningstar sustainability ratings affect flows: low-globe funds lose flows, while high-globe funds gain flows after ratings are introduced. But the response does not vary meaningfully with fund horizon.

Economic message. End investors care about ESG ratings on average, but investors in long-term funds do not appear uniquely more ESG-sensitive. This weakens the clientele-catering explanation.

Table XI

Fund Flows in Response to Morningstar Sustainability Ratings

This table examines how fund flows respond to the initial release of Morningstar sustainability ratings in March 2016. Fund-months between March 2015 and January 2017 are included in the sample. *Post rating* defines the period between March 2016 and Jan 2017. *Long-term fund* indicates mutual funds whose portfolio churn ratio is ranked at the bottom tercile as of March 2016. *Fractional rank of fund portfolio ESG score* is a continuous variable between zero and one, where a higher fractional rank indicates that the portfolio-average MSCI ESG score of a fund's holding ranks higher in the cross section. Standard errors, shown in parentheses, are clustered at the fund level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable: Monthly Fund Flows (%)	(1)	(2)
*One globe * Post rating*	−0.679**	−0.742**
	(0.292)	(0.367)
*One globe * Post rating * Long-term fund*		0.0960
		(0.509)
*Two globes * Post rating*	−0.0344	0.0347
	(0.226)	(0.276)
*Two globes * Post rating * Long-term fund*		−0.168
		(0.389)
*Four globes * Post rating*	0.313	0.0356
	(0.199)	(0.276)
*Four globes * Post rating* * Long-term fund		0.538
		(0.358)
*Five globes * Post rating*	0.744***	0.547
	(0.255)	(0.354)
*Five globes * Post rating * Long-term fund*		0.289
		(0.429)
Fractional rank of fund portfolio ESG score	0.225	0.392
	(0.913)	(0.919)
*Fractional rank of fund portfolio ESG score * Post rating*	−0.344	−0.622
	(0.723)	(0.727)
Ln(Fund lagged TNA)	−4.941***	−4.949***
	(0.830)	(0.830)
Fund Turnover Ratio	−0.759*	−0.825*
	(0.457)	(0.459)
Fund Expense Ratio	−1.026	−1.207
	(1.288)	(1.287)
*Post rating * Long-term fund*		0.611***
		(0.225)
Observations	21,856	21,856
Adjusted R^2	0.170	0.171
Fund FE	Y	Y
Investment Objective-by-Quarter FE	Y	Y

Table XII

Portfolio ESG Score and Investor Horizon: The Role of Investor Climate Sentiment

This table examines the relationship between investor horizon and portfolio ESG score in the context of investors' fluctuating sentiment about climate concerns. The dependent variable is the value-weighted average ESG score of a fund's portfolio holdings. Investor horizon is measured by turnover ratio, estimated fund churn ratio, or adjusted churn ratio. The Climate Concern Index is constructed by Engle et al. (2020) using climate change-related news articles. The fractional 12-month return rank for funds in the bottom within-objective quintile (*Low*) is defined as $Min(Rank_{t-1}, 0.2)$. Funds in the three medium return quintiles (*Mid*) are grouped together and given ranks according to $Min(0.6, Rank_{t-1} - Low)$. The rank for the top return quintile (*High*) is defined as $Rank_{t-1} - Mid - Low$. Investment objective-by-quarter fixed effects are included across specifications. Standard errors, shown in parentheses, are double-clustered at the fund and quarter level. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively. The sample period is 2000 to 2018.

Dependent Variable: Fund Portfolio Average ESG Score	(1)	(2)	(3)
Fund Turnover Ratio	−0.0947***		
	(0.0213)		
Fund Turnover Ratio * Climate Concern Index	0.00242*		
	(0.00122)		
Fund Churn Ratio		−0.205***	
		(0.0455)	
Fund Churn Ratio * Climate Concern Index		0.00378	
		(0.00292)	
Adjusted Churn Ratio			−0.526***
			(0.108)
Adjusted Churn Ratio * Climate Concern Index			0.00807
			(0.00679)
Holdings Ln(Market Cap)	0.644***	0.642***	0.643***
	(0.0511)	(0.0511)	(0.0511)
Holdings Book-to-Market	−0.218	−0.211	−0.212
	(0.179)	(0.178)	(0.179)
Holdings 12-Month Return	−0.232*	−0.227	−0.193
	(0.137)	(0.138)	(0.139)
Ln(Fund TNA)	−0.00637	−0.00648	−0.00620
	(0.00612)	(0.00614)	(0.00604)
Number of Portfolio Stocks (in 100s)	−0.0391***	−0.0422***	−0.0405***
	(0.00821)	(0.00822)	(0.00812)
Fund Return Rank – Low	−0.105	−0.111	−0.121
	(0.167)	(0.166)	(0.167)
Fund Return Rank – Mid	−0.0478	−0.0424	−0.0505
	(0.0659)	(0.0660)	(0.0661)
Fund Return Rank – High	−0.703**	−0.691**	−0.739***
	(0.273)	(0.273)	(0.274)
Observations	87,761	87,761	87,761
Adjusted R^2	0.825	0.825	0.825
Investment Objective-by-Quarter FE	Y	Y	Y

5.12 Table XII: climate-news sentiment

Read: Table XII interacts fund horizon with the Climate Change News Index. The interaction terms are close to zero or inconsistent with the catering prediction.

Economic message. Long-term funds do not increase their ESG tilt more when climate news becomes more salient. This provides another piece of evidence against the clientele-catering channel.

6 Short summary

- Long-horizon mutual funds and 13f institutions hold higher-ESG portfolios.
- High-ESG firms have longer-horizon institutional shareholder bases.
- The result is robust to excluding explicit ESG funds, adding style controls, and using alternative ESG ratings.
- Long-term funds are more likely to mention ESG considerations in shareholder reports.
- Long-term funds are patient after earnings misses but sell after negative environmental or social incidents.
- Funds with higher flow-performance sensitivity hold lower-ESG portfolios.
- The SEC 2004 disclosure shock suggests that shortening fund horizons reduces ESG tilts.
- The evidence does not support a simple catering story based on differential ESG preferences of long-term funds' end investors.

7 Conclusion

The paper shows that institutional ESG investing is closely connected to investor horizon. Long-term investors do not merely hold different stocks because of size, style, or explicit ESG mandates. They tilt toward higher-ESG firms, discuss ESG more often, react differently to ESG-related news, and reduce ESG exposure when regulatory changes shorten their horizon.

The economic interpretation is that ESG information is partly long-horizon information. Investors with longer horizons are better positioned to process it and to wait for its payoff. Short-horizon investors, especially those facing strong flow-performance pressure, may not be able to hold such positions even if they understand the long-run logic.

The final takeaway is that investor horizon is a central organizing variable for ESG ownership. ESG preferences are heterogeneous across institutions, and that heterogeneity is shaped by information processing, arbitrage limits, and flow pressure more than by simple clientele catering.